

Unit - 1

- Q1.** a. Consider two benchmark machines, with instruction set of 300,000 instructions, **3+2**
CO1 both have a clock rate of 500MHz. **= 5**

The following measurements are recorded on the two machines running a given set of benchmark programs:

Instruction Type	Instruction Mix	CPI
Machine A		
Arithmetic and logic	40	1
Data Transfer	25	2
Floating Point	20	4
Control Transfer	15	3
Machine B		
Arithmetic and logic	35	1
Data Transfer	20	3
Floating Point	15	4
Control Transfer	30	2

Explain and Determine the effective CPI, MIPS rate and execution time for both the machines.

Which of the machine will perform better and why?

- b. What do you understand by Von Neumann computation? Explain the organization using of a Von Neumann computer using a schematic diagram. Point out the shortcoming of the Von Neumann Architecture.

- Q2.** a. What must be the address field of an indexed addressing mode instruction be to **2+3**
CO1 make it same as a register indirect mode instruction? **= 5**

- b. An instruction is located at the location 4500 with its addressing field at locations 4501. The address field has value 250. A processor register R1 contains the number of 400. Evaluate the Effective Address if the addressing mode of the instruction is
- Direct Addressing Mode
 - Immediate Addressing Mode
 - Relative Addressing Mode
 - Register Direct Addressing mode
 - Index with R1 as a two digit hexadecimal number.

- (c) Define the following :
- Micro-operations
 - Micro-instruction
 - Micro-program

[1.5 *3= 4.5]

Q1.
CO1

- (a) A benchmark program is run first on a 200 MHz and then on a 400 MHz processor. The executed program consists of 1 million instruction execution with the following instruction mix and clock cycle count.

[4.5]

Instruction Type	Instruction Count	Cycles per instruction
ALU	4,50,000	1
Load and Store	2,50,000	2
Branching	1,50,000	3
Floating Point	1,00,000	4
Others	50,000	2

Calculate the effective CPI and MIPS rate for both the cases.

- (b) Name various types of CPU Organization. Discuss briefly the features of any one with the help of suitable diagram?
- (c) What are the various phases of instruction cycle? Explain with the help of micro-operations performed in each phase.

[4.5]

[4.5]

CO1

- (a) What is the primary (main) reason to study computer organization and architecture for a programmer?
- (a) What is the Von Neumann architecture? What is the unique feature of this architecture? What is the main drawback of this architecture? Explain in 3-5 sentences with a block diagram.
- (b) Consider two computers having model numbers A1 and A2 on which a 1000 instruction program is run. Computer A1 uses a 1 GHz clock and computer A2 uses a 2 GHz clock. The distribution of different types of instructions in the program and the average CPI for the two computers is shown in the following table. Determine the execution times of the program on the two machines.

Instruction Type	Percentage of Occurrence	CPI - Machine A1	CPI - Machine A2
Load/Store	10	5	10
Arithmetic	20	5	5
Other	70	5	8

Q2. (2+2) 5 Marks

Q2, (2+3) 5 Marks

CO1

- What are the three different types of computer instructions? How are they different with respect to their 4 bit opcodes?
- List the register transfer statements along with the necessary control functions for the instruction execution cycle including the stages for fetching, decoding and determining the instruction type.

Q3, (1+2+2) 5 Marks

CO1

- If a computer has a 16 bit wide memory with 4096 words, how many bits does an instruction have? How many bits are reserved for the address?
- Write register transfer statements including fetch and decode with control functions to add a memory word to the AC with direct addressing (op-code for ADD is 0, I = 0 implies direct addressing).
- Write register transfer statements including fetch and decode with control functions to add a memory word to the AC with indirect addressing (op-code for ADD is 0, I = 1 implies indirect addressing).

Q1.
CO1

- (a) We want to compare computers R1 & R2. Both have a clock frequency 400 MHz. We run the same program on both, which has the following mixture of instructions: [1.5*3 = 4.5]

Type of Inst.	%	R1 CPI	R2 CPI
FP add	16% <i>x, y</i>	6	20
FP multiply	10% <i>0, b</i>	8	32
FP divide	8% <i>w, z</i>	10	66
Non-FP	66%	3	3

- Calculate the MIPS for the computers R1 and R2.
- Calculate the CPU program execution time on the computers R1 and R2, if there are 12000 instructions in the program?
- At what mixture of instructions will the computers be equally fast?

- (b) The DEC PDP-11 minicomputer had an 18-bit address. Answer the following questions: [1.5*3 = 4.5]

- What was the largest possible memory in bytes if a memory location is 1 Byte?
- How long should the program counter (PC) of the computer be?
- What would need to change in the computer if we would like to increase address space 8-times?

- (c) Draw the block diagram for the hardware that implements the following register transfer along with control function: [4.5]

$$x + yz : AR \leftarrow AR + BR$$

Where x, y and z are control signals and AR and BR are registers. Note that when "+" is used in the control function, it means OR and when it is used on the right hand side, it means addition.

Q1 (2+3) 5 Marks CO1

- (a) What is the Von Neumann architecture? What is the unique feature of this architecture? What is the main drawback of this architecture? Explain in 3-5 sentences with a block diagram.
- (b) Consider two computers having model numbers A1 and A2 on which a 1000 instruction program is run. Computer A1 uses a 1 GHz clock and computer A2 uses a 2 GHz clock. The distribution of different types of instructions in the program and the average CPI for the two computers is shown in the following table. Determine the execution times of the program on the two machines.

Instruction Type	% of Occurrence	CPI - Machine A1	CPI - Machine A2
Load/Store	20	8	12
Arithmetic	20	5	5
Other	60	5	8

Q2 (3+2) 5 Marks**CO1, CO2**

- (a) What are the three different types of computer instructions? How are they different with respect to their 4 bit opcodes?
- (b) Let's say we have three control outputs, x, y and z. Draw a circuit that executes the register transfer. $x \text{ OR } y \text{ OR } z : R1 \leftarrow R2$

Unit - 2

Q2 5 Marks

CO2

Draw a block diagram of a hardwired control unit. Explain its parts, functions and operation. What does the following register transfer accomplish? (Op-code 1 means ADD)

$D_1T_4 : DR \leftarrow M[AR]$

$D_1T_5 : AC \leftarrow AC + DR$

Q1. Given that a simple CPU can execute the following instructions:

5

CO2

Instruction	Instruction Code	Operation
ADD	00AAAAAA	$AC \leftarrow AC + M[AAAAAA]$
JMP	10AAAAAA	GOTO AAAAAA
INR	11XXXXXX	$AC \leftarrow AC + 1$

Design the hardwired control for the above instructions.

(a)

Given that a simple CPU can execute the following instructions:

[4.5]

Instruction	Instruction Code	Operation
JMP	00AAAAAA	$PC \leftarrow AAAAAA + 1$
SUB	01AAAAAA	$AC \leftarrow AC - M[AAAAAA]$
ADD	10AAAAAA	$AC \leftarrow AC + M[AAAAAA]$
SKIP	11XXXXXX	$PC \leftarrow PC + 1$

Design the hardwired control for the above instructions.

Q2, (1.5+1+1+1.5) 5 Marks

CO2

- What are the two main functions of a microprogrammed control unit? Define "microinstruction."
- Which register is used to hold return addresses in microprogrammed control?
- The last microinstruction in the microprogram for a **STORE** instruction is **WRITE U JMP FETCH**. Fully specify the symbolic form of the microinstruction that is executed immediately after this one.
- Provide 3 justifications for storing microprograms in a ROM.

Unit - 3

Q2 (1+2+2) 5 Marks

CO2

- A. What are the two main functions of a microprogrammed control unit? Define "control word" and give an example of a 20-bit control word format. Explain how the bits are used.
- B. The last microinstruction in the microprogram for a **STORE** instruction is **WRITE U JMP FETCH**. Fully specify the symbolic form of the microinstruction that is executed immediately after this one.
- C. Provide 3 justifications for storing microprograms in a ROM.

Q3 5 Marks

CO2

Explain the operation of a microprogram sequencer with the help of a block diagram.

Q1'. Explain the significance of horizontal and vertical microinstruction with example.
CO2

(b) Draw the diagram of a generic micro-program sequencer.

[4.5]

Q3, 5 Marks (Attempt any one part)

CO2/CO3

- A. Explain the operation of a microprogram sequencer with the help of a block diagram.

OR

- B. A set-associative cache has a set size of 4 words. A least recently used (LRU) replacement algorithm is used. A value from 0 to 3 is recorded for each word. When a hit occurs, the counter associated with the referenced word is set to 0. Counters with values originally lower than the referenced one are incremented by 1, all others are unchanged. If a miss occurs, the word with counter value 3 is removed, the new word is put in its place and its counter is set to 0. The other three counters are incremented by 1. Show that the procedure works for the sequence of word references: A, B, C, D, B, E, D, A, C, E, C, E (Start with A,B,C,D as the initial four words with word A being the least recently used).

Q2. (a)
CO2

A microprogrammed control organization has the following delays: 40 ns to generate the next address, 10 ns to transfer the address into the CAR, 40 ns to read the control memory ROM, 10 ns to transfer the microinstruction to the

[4.5]

(b)

control data register and 40 ns to execute the microoperations specified by the control word. What is the maximum clock frequency that the control can use? What would the clock frequency be if the control data register is not used? If we have the following microprogram for the ADD instruction, what should be the microprogram for INDRCT2?

[4.5]

ADD: READ I CALL INDRCT2
ADD U JMP FETCH

(c)

Define the following :

- (i) Micro-operations
- (ii) Micro-instruction
- (iii) Micro-program

[2.5 + 2 = 4.5]

Show how a 9-bit micro-operation field can be divided to encode 46 microoperations. How many micro-operations can be specified in each micro-instruction?

Unit - 4

Q1 (3+2) 5 Marks

CO3

- A. Draw a schematic diagram of a memory hierarchy. What is the primary reason to use a memory hierarchy instead of a single large memory? Explain.
- B. Explain the principle of *locality of reference*.

Q1 (CO3)	Consider a Main Memory Size = 64 GB, Cache Size = 128 MB and Block Size = 1KB. For a direct-mapped cache, a main memory address is viewed as consisting of three fields. List and define the three fields with value of each. What is the drawback of using Direct-mapped cache? Explain it.	5
Q2 (CO3)	Define and give example: (i) Principal of locality. (ii) Register. (iii) Memory Hierarchy.	6
Q3 (CO3)	Design Static RAM cell and explain its read and write operation in the SRAM.	4

Q3.
CO3

- a. For a direct mapped cache a main memory address is viewed as consisting of different fields. Using Hardware implementation diagram define the use of fields in the direct Mapping.
- b. A direct Mapping cache consist of Main memory Size of 64 GB and the size of cache is 512 KB with each line size consists of 256B. Calculate the Physical Address Length, No of lines in Cache and Tag bits.

2.5
+
2.5
= 5

Q2.
CO3

- a. What is difference between Write-through and Write-back cache?
- b. Explain the term locality of reference? How is it used to improve the performance of cache memory?

2+3
= 5

- Q3. (a) (i) A set-associative cache consists of 64 lines, or slots, divide into two lines sets. The main memory contains 4K blocks of 128 words each. Show the format main memory address. [2.5 + 2 = 4.5]
CO3
- (ii) Consider a Main Memory Size = 256 GB, Cache Size = 512 MB and Block Size = 16 KB.
1. What would be the size of Tag Bits if it is 4-way set associative memory mapping.
 2. Find No of Sets and comparators required in the mapping.
- (b) Why a block replacement algorithm is required in cache mapping. Name all such algorithms. Define any two of them. [4.5]
- (c) Write short notes on: [1.5 * 3 = 4.5]
- (i) SRAM and DRAM
 - (ii) Memory Management Unit (MMU).
 - (iii) Interleaved Memory

- Q3. (a) A set-associative cache has a set size of 4 words. A least recently used (LRU) replacement algorithm is used. A value from 0 to 3 is recorded for each word. When a hit occurs, the counter associated with the referenced word is set to 0. Counters with values originally lower than the referenced one are incremented by 1, all others are unchanged. If a miss occurs, the word with counter value 3 is removed, the new word is put in its place and its counter is set to 0. The other three counters are incremented by 1. Show that the procedure works for the sequence of word references: A, B, C, D, B, E, D, A, C, E, C, E (Start with A,B,C,D as the initial four words with word A being the least recently used). [4.5]
CO3
- (b) A computer has a memory of 64K words each of size 1 byte and a direct mapping cache with 1024 words each of size 1 byte and blocks consisting of eight words each. [1.5*3 = 4.5]
- (i) How many bits does a memory address have?
 - (ii) Write the address format for the cache showing the number of bits for index, tag, block and word offset.
 - (iii) If block size were reduced to 1 word, how would cache performance be affected?
- (c) Write short notes on: [2.5+2 = 4.5]
- (i) Least Recently Used versus FIFO replacement algorithm
 - (ii) Write-back versus write-through policies

Unit - 5

Q3. a. Design a parallel priority interrupt hardware for a system with eight interrupt sources and explain its working? **2+3 = 5**
CO4

b. List and Explain PCI/PCIe Protocol layers? What is PCIe multilane distribution?

Q4. (a) What is the difference between isolated I/O and memory-mapped I/O? What are the advantages and disadvantages of each? **[4.5]**
CO4

- (b)
- (i) What are the functions of an I/O interface? Explain. **[1.5 * 3 = 4.5]**
 - (ii) Draw a block diagram of I/O interface showing the major components.
 - (iii) What is the function of status and control registers?

(c) Why do we require Direct Memory transfer? Draw the diagram showing the connections of DMA controller with the CPU and I/O devices. What are the steps performed in a DMA transfer? **[4.5]**

Q4. (a) Show the organization and functioning of a parallel priority interrupt encoder with the help of a block diagram and truth table. **[4.5]**
CO4

3/8/22 Field num

- (b)
- (i) Give four reasons for using an I/O interface. **[1.5 * 3 = 4.5]**
 - (ii) Draw a block diagram of I/O interface showing the major components.
 - (iii) What is the function of status and control registers?

(c) Why do we require Direct Memory transfer? Draw the diagram showing the connections of DMA controller with the CPU and I/O devices. What are the steps performed in a DMA transfer? **[1.5 * 3 = 4.5]**